Monge-Ampère Equations—Bernstein Problem*

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An ellipsoid centered at the origin can be written as

$$E(A) = \{x \in \mathbb{R}^n \mid \langle Ax, x\rangle < 1\},$$

for some positive definite matrix A.Here,$\langle\cdot,\cdot\rangle$ is the standard inner product in $\mathbb{R}^n$.The volume of $E(A)$ is given by

$$|E(A)| = \frac{|B_1|}{\sqrt{\det(A)}}.$$

We always write,for any subset $S \subset \mathbb{R}^n$ and any $\tau > 0$,

$$\tau S = \{\tau x \mid x \in S\}.$$

LEMMA 0.1 (John Lemma) *Let Ω be a bounded convex domain in $\mathbb{R}^n$,with $0 \in \Omega$ as the center of mass of Ω.Then,there exists an ellipsoid E with its center at the origin such that*

$$n^{-3/2}E \subset \Omega \subset E.$$

Moreover,E can be taken as the ellipsoid of minimum volume containing Ω with its center at the origin.

We always write

$$\tau_n = n^{-3/2}.$$

A convex domain Ω is **normalized** if $0 \in \Omega$ is the center of mass of Ω and

$$B_{\tau_n} \subset \Omega \subset B_1.$$

LEMMA 0.2 *Let Ω be a normalized convex domain in $\mathbb{R}^n$.Suppose that u is a nonnegative convex $C(\bar{\Omega})\cap C^4(\Omega$ solution of*

$$\begin{aligned} \det D^2u &= 1 \quad \text{in } \Omega, \\ u &= h \quad \text{on } \partial\Omega, \\ u(x_0) &= 0 \quad \text{for some } x_0 \in \Omega, \end{aligned}$$

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for some positive constant h.Then,
(1)there exist two positive constants c_1 and c_2,depending only on n,such that

$$c_1 \leq h \leq c_2.$$

(2)there exists a constant $\tau_0 \in (0,1)$,depending only on n,such that $B_{\tau_0}(x_0) \subset \Omega$ and

$$u < \frac{1}{2}h \quad \text{in } B_{\tau_0}(x_0).$$

(3)there exists a constant $\alpha \in (0,1)$,depending only on n,such that

$$C_1 I \leq D^2 u \leq C_2 I \quad \text{in } S_{h/2}$$

and

$$[D^2 u]_{C^\alpha(B_{\tau_0}(x_0))} \leq C,$$

where $S_{h/2} = \{x \in \Omega \mid u(x) < \frac{h}{2}\}$,$C_1, C_2$ and C are positive constants depending only on n.

We now prove an important result on the interior Hölder semi-norms of the second derivatives.

THEOREM 1 *Let Ω be a bounded convex domain in $\mathbb{R}^n$.Suppose that u is a nonnegative convex $C(\bar{\Omega}) \cap C^4(\Omega)$ solution of*

$$\begin{aligned} \det D^2 u &= 1 \quad \text{in } \Omega, \\ u &= 1 \quad \text{on } \partial\Omega. \end{aligned}$$

Assume $0 \in \Omega$ such that

$$u(0) = 0, \qquad D^2 u(0) = I.$$

Then,there exist constants $\mu > 0$ and $\alpha \in (0,1)$,depending only on n,such that $B_\mu \subset \Omega$ and

$$[D^2 u]_{C^\alpha(B_\mu)} \leq C,$$

where C is a positive constant depending only on n.

PROOF Let $x_0 = (x_{0,1}, \cdots, x_{0,n}) \in \mathbb{R}^n$ be the center of mass of Ω and E be the ellipsoid of minimum volume containing Ω with its center at x_0.
By taking an appropriate rotation O,we assume that the axes of the ellipsoid E are parallel to the coordinate axes and that E is given by

$$E = \left\{ x \mid \sum_{i=1}^{n} \frac{(x_i - x_{0,i})^2}{a_i^2} < 1 \right\},$$

for some positive constants $a_1, \cdots, a_n$.
Consider the matrix

$$A = \operatorname{diag}\left(\frac{1}{a_1}, \cdots, \frac{1}{a_n}\right)$$

and the affine transform

$$y = Tx = A(x - x_0).$$

We have

$$T(x) = A(x - x_0) = \begin{pmatrix} \frac{1}{a_1} & & \\ & \ddots & \\ & & \frac{1}{a_n} \end{pmatrix} \begin{pmatrix} x_1 - x_{0,1} \\ \vdots \\ x_n - x_{0,n} \end{pmatrix} = \begin{pmatrix} \frac{x_1 - x_{0,1}}{a_1} \\ \vdots \\ \frac{x_n - x_{0,n}}{a_n} \end{pmatrix}$$

Since$\sum_{i=1}^{n} \left(\frac{x_i - x_{0,i}}{a_i} \right)^2 < 1$,then

$$T(E) = B.$$

By **John Lemma0.1**,we have

$$B_{\tau_n} \subset T(\Omega) \subset B_1.$$

Set $v(y) = hu(x)$ for any $y \in T(\Omega)$,where h is chosen such that $h^n(\det A^{-1})^2 = 1$.Since

$$D_i v(y) = D_i(hu(x)) = D_i(hu(A^{-1}y + x_0)) = hD_i u(x)A^{-1}$$

and

$$D_{ij} v(y) = hD_{ij} u(x)(A^{-1})^2,$$

then

$$\det D^2 v(y) = \det(hD^2 u(x)(A^{-1})^2) = h^n \det(D_{ij} u) \det(A^{-1})^2 = 1.$$

The function v is convex and satisfies

$$\begin{aligned} \det D^2 v &= 1 \quad \text{in } T(\Omega), \\ v &= h \quad \text{on } \partial(T(\Omega)), \\ v(y_0) &= 0, \end{aligned}$$

where $y_0 = -Ax_0 \in T(\Omega)$.
Apply **Lemma0.2** to v,we have

- $$c_1 \le h \le c_2,$$

where c_1 and c_2 are positive constants depending only on n.

- there exists a positive constant τ_0,depending only on n,such that

$$B_{\tau_0}(y_0) \subset \left\{ y \in T(\Omega) \mid v(y) < \frac{h}{2} \right\}.$$

- $$C_1 I \le D^2 v \le C_2 I \quad \text{in } B_{\tau_0}(y_0),$$

where C_1 and C_2 are positive constants depending only on n,and

$$[D^2 v]_{C^\alpha(B_{\tau_0(y_0)})} \le C,$$

where $C > 0$ and $\alpha \in (0, 1)$ are constants depending only on n.
Since

$$D^2 v(y_0) = h(A^{-1})' D^2 u(0) A^{-1} = h(A^{-1})' A^{-1},$$

then,

$$C_1 I \leq h(A^{-1})' A^{-1} \leq C_2 I.$$

Since $c_1 \leq h \leq c_2$,then,

$$C_1 \leq a_i \leq C_2,$$

and hence

$$T(B_{c_0 \tau_0}) \subset B_{\tau_0}(y_0),$$

for some positive constant c_0 depending only on n.
It is obvious that

$$v_i(y) = (hu(A^{-1}y + x_0))_i = hu_1 a_i,$$

and

$$v_{ij}(y) = (hu_i(A^{-1}y + x_0)a_i)_j = ha_i a_j u_{ij}.$$

Then,

$$u_{ij} = \frac{1}{ha_i a_j} v_{ij}(y),$$

and

$$\frac{v_{ij}(y)}{c'} \leq u_{ij} \leq \frac{v_{ij}(y)}{c''}.$$

Since

$$[D^2 v]_{C^\alpha(B_{\tau_0}(y_0))} \leq C,$$

thus

$$[D^2 u]_{C^\alpha(B_{c_0 \tau_0})} \leq C.$$

□

THEOREM 2 (Bernstein Problem) *Suppose that u is a convex $C^4(\mathbb{R}^n)$ solution of*

$$\det D^2 u = 1 \quad \text{in } \mathbb{R}^n.$$

Then,u is a quadratic polynomial.

PROOF Without loss of generality,we assume

$$u(0) = 0, \quad Du(0) = 0, \quad D^2 u(0) = I.$$

By the convexity,u assumes its minimum at 0 and hence $u \geq 0$.
For an arbitrary fixed constant $R > 0$,we set

$$\Omega = \{x \in \mathbb{R}^n \mid u(x) < R^2\}.$$

Set $x = Ry$ and $v(y) = \dfrac{1}{R^2} u(x)$.
Set

$$\widetilde{\Omega} = \{y \in \mathbb{R}^n \mid x = Ry \in \Omega\}.$$

We have

$$D_i v(y) = D_i \left(\frac{1}{R^2} u(Ry) \right) = \frac{1}{R} D_i u(x),$$

and

$$D_{ij}v(y) = D_j\left(\frac{1}{R}D_i u(Ry)\right) = D_{ij}u(x).$$

Thus

$$\begin{aligned} \det D^2 v &= 1 \quad \text{in } \widetilde{\Omega}, \\ v &= 1 \quad \text{on } \partial\widetilde{\Omega}, \\ v(0) &= \min_{\widetilde{\Omega}} v = 0 \\ D^v(0) &= 1. \end{aligned}$$

By **theorem1**,$B_\mu \subset \widetilde{\Omega}$ and

$$[D^2 v]_{C^\alpha(B_\mu)} \leq C,$$

where $\mu > 0$,$\alpha \in (0,1)$,and $C > 0$ are constants depending only on n.Back to u,we have $B_{\mu R} \subset \Omega$ and

$$R^\alpha [D^2 u]_{C^\alpha(B_{\mu R})} \leq C.$$

Letting $R \longrightarrow \infty$,we conclude that $D^2 u$ is constant in $\mathbb{R}^n$. □

References

[1] Qing Han, Nonlinear Elliptic Equations of the Second Order, American Mathematical Society, GSM171, 2016.